

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 06)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

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Evaluate the success of the methods and strategies used to reduce global climate change.

[9 marks]

Extra space

0 1 . **2** State what is meant by a **carbon footprint**.

[1 mark]

10

0 2 Table 1 shows the mean surface temperature of the Earth with and without an atmosphere containing greenhouse gases.

Table 1

Condition	Mean surface temperature / °C
No greenhouse effect	-18
Natural greenhouse effect	15

0 2 . **1** Use Table 1 to calculate the warming produced by the natural greenhouse effect.

[1 mark]

Answer

°C

0	2	.	2
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Explain how the natural greenhouse effect warms the Earth's surface.

[4 marks]

0	2	.	3
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Explain what is meant by a **greenhouse gas**.

[1 mark]

0	2	.	4
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The wavelengths of radiation arriving at the Earth are different from those leaving. Describe this difference and explain why it matters to the energy budget.

[2 marks]

0	2	.	5
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Explain why a change in the movement of energy through the atmosphere could affect the distribution of species.

[2 marks]

10

03 . 1

Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

- A Aerobic decomposition produces little methane and returns nutrients ☐ to the soil
- B It produces more electricity ☐
- C It removes heavy metals ☐
- D It destroys asbestos fibres ☐

03 . 2

Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

03 . 3

Give **two** features of a well-managed landfill site that reduce pollution.

[2 marks]

1

2

5

Neonicotinoids are the most widely used group of insecticides. Explain how the properties of neonicotinoids affect their impact on bee populations and describe the methods used to reduce this impact.

ESP/7447/1/SET06

Extra space

0 4 . 2 State **one** way in which pesticide use in agriculture may be reduced by using natural processes.

[1 mark]

10

Complete Table 1 to show a suitable sampling technique for each group of organisms and one limitation of the technique.

Do not write outside

Organisms	Technique	One limitation	
Invertebrates active on the ground surface at night			no
Invertebrates living in leaf litter			
Invertebrates on tree branches			

Explain why a light trap cannot be used to prove that a moth species is absent from a site.

Explain how a motor-driven suction sampler can give a population density, whereas a pooter cannot.

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10

06

Table 1 shows the mean daily solar energy received on a horizontal surface at three locations.

Table 1

Location	Latitude	Mean solar energy in June / kWh m ⁻² day ⁻¹	Mean solar energy in December / kWh m ⁻² day ⁻¹
Sahara desert	25°N	7.8	4.2
Southern England	51°N	5.0	0.7
Northern Norway	69°N	4.6	0.0

06

. 1

Use Table 1 to calculate how many times more solar energy is received in December in the Sahara than in southern England.

[1 mark]

Answer

times

06

. 2

Explain why northern Norway receives more solar energy in June than in December.

[2 marks]

06

. 3

Explain why concentrating solar power systems using parabolic reflectors only work where there is no cloud.

[2 marks]

06 . 4

Explain why the low energy density of solar power is a problem for large-scale generation.

[1 mark]

06 . 5

Evaluate the extent to which new technologies could overcome the problems of using solar power as a large-scale energy resource.

[9 marks]

Do not write outside the box

Extra space

15

0 7 . 1

Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]A High sand content ☐B High clay content ☐C Crumb peds ☐D High pH ☐**0 7 . 2**

Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

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0 7 . 3

Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

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5**0 8**

A factory discharges a persistent, water-soluble pollutant into a river. Table 1 shows the concentration of the pollutant measured at sites along the river on a day with normal river flow.

Table 1

Site	Distance downstream of discharge / km	Pollutant concentration / mg dm ⁻³	River flow / m ³ s ⁻¹
A (upstream)	-2	0.0	12
B	0.5	4.8	12
C	5	3.9	15
D (below tributary)	12	1.6	36
E	30	1.4	38

08 . 1

Calculate the percentage decrease in pollutant concentration between site B and site E.

[2 marks]

Answer

%

08 . 2

Explain the large decrease in concentration between sites C and D.

[2 marks]

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08 . 3

Explain why the concentration changes very little between sites D and E.

[2 marks]

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08 . 4

Suggest why the pollution might be more serious if the same discharge entered a lake instead of a river.

[2 marks]

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08 . 5

Explain why the factory might be permitted to discharge the effluent into the sea near a strong current but not into the river.

[2 marks]

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10

09 . 1 Describe **three** effects of noise on human health other than hearing damage. **[3 marks]**

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2

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09 . 2 Explain how noise can damage non-living objects. **[3 marks]**

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09 . 3 Explain how noise can reduce the breeding success of birds. **[2 marks]**

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09 . 4 Explain why military sonar is a threat to cetaceans. **[2 marks]**

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10

10 . 1

Discuss how the features of energy resources determine their suitability for different uses and their environmental sustainability.

[9 marks]

Extra space

10 . **2** State **one** reason why nuclear power stations can be located where transport infrastructure is poor.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1}\boxed{1}.\boxed{1}$$

Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

OR

1 1 . 2

Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS